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Eco-SMART – A Unified and Scalable Cloud-Computing Platform for Seamlessly Integrating Mental Health and Wellness Apps and Services

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Abstract: Mental health and wellness (MHW) refer to the emotional, psychological, and social well-being of individuals. It has become a global challenge to the proper functioning of societies because psychological disorders have become major contributors to social impediments, financial burdens, and individual disabilities, costing trillions of dollars in socio-economic losses. Incidentally, such technologies as artificial intelligence and real-time big data analytics offer immense promise in advancing MHW detection, treatment, and management. Consequently, the role of technology in MHW is growing exponentially and the ubiquity of smartphones makes mobile apps worthwhile tools in delivering MHW related interventions. However, there is a lack of unified and scalable platforms for supporting such domain-specific MHW apps. Such platforms could provide a variety of requisite services and capabilities to MHW app developers including multimodal MHW data acquisition, data processing, secure data storage, data authentication, marker detection, large language model (LLM) orchestration, triangulation, interpretation, intelligent analytics, and other APIs. Such a unified, scalable, intelligent, and adaptable platform simplifies the process of connecting an app with these services, this enables MHW app developers to focus on building the core functionality rather than dealing with complex backend infrastructure. In addition, such a platform facilities seamless integration and interoperability of various MHW apps. In this paper, we propose system architecture for a comprehensive platform for plugging-in and integrating MHW apps and services in a unified, scalable, adaptive, and extensible manner. This initiative would spur the growth of MHW apps by providing effective and productive app design and development services and support. Furthermore, it would help promote the proliferation and acceptance of MHW apps by efficacious support to users and caregivers.

Keywords: Holistic Mental Wellness; AI-based Adaptive Platform; Application Programming Interface (API); Continuous Integration/Continuous Deployment (CI/CD); Containerization Platforms; Application Deployment and Hosting; Application Analytics and Reporting

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1. Introduction

Mental health and wellness (MHW) describes the emotional and psychological and social health status of people in their communities along with groups [1]. MHW acts as a critical foundation for all areas of human existence that includes understanding and intentional behavior and personal life problem solving ability [2], [3]. Indeed, MHW has become a global challenge to the proper functioning of societies with psychological disorders becoming major contributors to social impediments, financial burdens, and individual disabilities [4], [5]. A non-exhaustive list of such common disorders would include Major Depressive Disorder (MDD), Post-Traumatic Stress Disorder (PTSD), obsessive-compulsive disorder (OCD), Generalized Anxiety Disorder (GAD), Social Anxiety Disorder (SAD), Bipolar Personality Disorder (BPD), Addiction / Substance Abuse, etc. Multiple factors including fast urban growth together with technological changes and economic differences and international conflicts and health emergencies have intensified MHW issues. Problems in MHW go unchecked because they produce extreme consequences which lead to individual misery as well as reduced life quality and reduced productivity and elevated health system costs and relationship problems and sometimes even death [1]. Globally MHW challenges are both multiplying and becoming more difficult to understand at a faster rate.

MHW previously operated under social misconceptions and prejudices so many individuals avoided help and discredited its role in life quality improvement [6], [7], [8]. In recent times both the UNO and numerous jurisdictions have elevated mental wellness to basic human rights status [9]. For instance, the United Nations 2030 Agenda for Sustainable Development emphasizes well-being for all ages. The Goal-3 (SDG-3) makes healthy life promotion and well-being enhancement its main target UNO [10]. The United Nations recognizes mental health as an essential element in building sustainable development [11].

Indeed, an estimated 25% to 50% of the global population encounters a MHW issue during their lifetime [7]. As such, MHW is already deemed a global crisis, affecting hundreds of millions of lives and costing trillions of dollars in terms of socio-economic losses every year [12]. MHW care investments yield exceptionally high returns on investment. The treatment of depression and anxiety using every dollar creates a four-dollar benefit for enhanced health and better productivity levels [12]. The current assessment and intervention procedures for mental health rely primarily on clinical observations together with self-reported data but these methods tend to be objective and time-sensitive and adaptable. Inadequate healthcare methods used by MHW result in delayed diagnosis together with insufficient support and major social consequences.

In this regard, such technologies as artificial intelligence, big data analytics, and real-time analytics offer a huge potential in advancing MHW detection, monitoring, and management [[13], [14], [15], [16]]. Consequently, technology-enabled and -augmented solutions in MHW are receiving due attention at various levels and circles in society, industry, and government. Such technology-driven intelligent and knowledge-based approaches could provide multifaceted benefits at minimal cost while providing widespread availability. Such benefits would include proactive detection of MHW issues, continuous and real-time remote monitoring of subjects, objective and knowledge-based diagnosis and treatment planning. These benefits would also include holistic and multimodal decision-making, timely intervention, adaptive and personalized care / services, higher degree of accessibility and availability, ethically and socially sensitive information privacy and security, etc. Regular updates / upgrades, user feedback, and integration of the latest advancements in Artificial Intelligence (AI) and MHW research are also integral to technology-based approaches. The advantages of technology in MHW stand strong but professional MHW care requires different attention from technological solution [11]. The proper use of technology is needed to support the existing activities of qualified medical professionals when detecting and managing mental health concerns within clinics and institutions. AI technology becomes crucial because the high numbers of MHW-related deaths and insufficient MHW treatments both support its importance [17].

It should not come as a surprise that the possibilities of the role of technology in MHW have been also expanding exponentially in recent years. Regarding this, different digital tools and MHW related apps have originated as potential choices of traditional therapeutic interventions. Of course, AI is indeed a powerful and transformative force that could change the face of MHW domain. AI brings new highly promising paradigms for personalized care, understanding individual user nuances, presenting their needs and adjusting interventions in real time. Anytime AI works on processing immense amounts of user data and analyzing them to discover patterns, get insight to propose feasible solutions.

Mobile apps benefit from smartphone ubiquity because these devices give primary access to MHW related intervention tools. The present app stores feature more than 10,000 MHW applications. Research indicates that the worldwide digital MHW market will expand from USD 5.2 billion during 2022 to USD 25 billion in 2030 [18]. Various driving factors including growing public awareness together with pandemic stress and acceptance of technology contribute to this market expansion. The Federal Drug Administration has officially approved multiple application programs for MHW detection and management purposes. Digital solutions matter significantly to FDA when addressing MHW issues. Digital mental health solutions with apps receive endorsement from various governments because they want to improve mental health service accessibility through digital healthcare measures. The integration and acceptance magnitude of mental health apps differs between healthcare structures and national setups yet governments demonstrate increasing acceptance of their potential merits [19], [20].

Digital health initiatives from numerous governments actively use technology such as applications to boost healthcare service delivery. Initiatives under these programs support the creation and acceptance of mental health applications which form part of strategic digital health policies. Governments include mental health among their policy priorities while funding services for mental health through allocated budgets. Some elements of digital mental health solutions such as apps receive support through the allotted government funds that operate within broader strategies to enhance mental healthcare availability.

The essential function of policymakers together with healthcare organizations involves defining guidelines and supplying financial support as well as platform adoption within existing healthcare systems. Healthcare apps require approval from governmental bodies which also monitor their operation as part of their oversight responsibilities. The official recognition or recommendation of mental health apps depends on assessment conclusions from these regulatory bodies regarding safety together with effectiveness along with privacy protection of the apps.

Users can utilize the general-purpose platforms iOS and Android to develop apps using their basic features. The developing and deploying of apps receives extensive support from cloud platforms which include Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP). They provide scalable infrastructure, data storage, analytics, and enable developers to build robust and scalable apps. However, there is a dearth of platforms for supporting domain-specific MHW apps. Developers of MHW related apps must build all the requisite services and capabilities. These services would include multimodal MHW data acquisition, data processing, secure data storage, data authentication, MHW marker detection, MHW model orchestration, triangulation, interpretation, intelligent analytics, and APIs. Such a unified, scalable, intelligent, and adaptable platform simplifies the process of connecting an app with these services, enabling MHW app developers to focus on building the core functionality rather than dealing with complex backend infrastructure. In addition, such a platform also facilities seamless integration and interoperability of various MHW apps.

To address this need, we propose a system architecture system for a comprehensive platform for plugging-in and integrating MHW apps and services focused on creating an Eco-System for Mental Wellness and Adaptive Rehabilitation using Artificial Intelligence (Eco-SMART). Eco-SMART is a unified, scalable, and extensible platform supporting such apps. This initiative is rooted in the profound understanding and awareness of the pressing need for innovative, personalized, and adaptive solutions for addressing the MHW challenges posed to societies globally.

The rest of the paper is organized as follows. Section-2 provides a brief but encompassing literature survey on relevant domains. Section-3 gives detailed discussion of the Eco-SMART platform architecture and our vision regarding the working of various Eco-SMART layers and components. Section-4 outlines our vision regarding the working of various Eco-SMART within the context of MDD Case Study. Section-5 provides some interesting and rewarding future directions. Section-6 concludes the paper.

2. Literature Survey

This multifaceted work draws from research in AI-driven automated mental health assessment, screening, interventions, predictive modeling, risk assessment, emotion recognition, sentiment analysis, personalized treatment, digital phenotyping, behavioral monitoring, etc. In addition, it relies on cloud-computing platforms for development and deployment of applications handling sensitive data. Consequently, this survey of the relevant literature is inherently expansive and unconventional. Nevertheless, here we provide a quick review of literature highlighting the significance of MHW through multimodal triangulation approaches as well as challenges and opportunities within the context of web and mobile apps for MHW related activities.

2.1. Significance of Mental Health and Wellness

Here we provide a brief survey of literature underscoring the critical importance of MHW in society and its profound impact on individual and social well-being, public health, and global development. Indeed, MHW challenges are ubiquitous, as 25-50% of the global population gets affected by MHW issues at some point in their lives [7]. The situation worsens when COVID-19 emerges alongside natural disasters which drives more people to experience mental stress and disorders. Innovation becomes essential to bridge the existing gap in MHW related care. Current unforeseen global circumstances including pandemics make it clear that an effective mental health infrastructure is essential for populations. The United Nations 2030 Agenda for Sustainable Development maintains well-being as an essential element to benefit all ages equally. SDG-3 serves as a specific goal which aims to achieve healthy lives and promote mental health worldwide [10].

To describe Eco-SMART as an MHW app platform tangibly, we have framed our running discussions within an identifiable MHW issue, namely Major Depressive Disorder (MDD). In addition, Section-5 details a coherent case study on how an MDD related MHW app could benefit from our comprehensive Eco-SMART platform. The premise of selecting MDD for our case study is that MDD had a significant impact on the global productivity and economy. It is estimated that MDD causes a loss of 12 billion days each year globally, costing the world economy $1 trillion [[12]]. [21] provides a detailed review current state of mental health issues and highlights that the high prevalence of MHW issues in the general population [[21]]. For instance, the prevalence of depression or anxiety has exceeded 12% of the population.

2.2. Significance of Multimodal Paradigm in MHW Applications

Several researchers adopt Multimodal paradigm to merge multiple data types for analytical purposes [22]; [23]. Text, images, audio, video, as well as other sensory physiological and behavioral inputs form the modes covered by the multimodal paradigm [24];

[25]. Research under the multimodal paradigm framework pursues fresh understanding that emerges through integrating several data types together [13], [26].

The MDD case requires quantitative or numerical data points to deliver valuable statistical information regarding subject mental state [27]. Different quantitative data collection methods could include Structured Surveys and Self-Reports using the PHQ-9 as well as Physiological Data measurement of heart rate and sleep patterns and cortisol levels together with Social Media Analytics of specific keyword frequency and patterns and post and response time tracking and Audio/Voice Data Analysis of speech rate and tone and pitch and Video Data Analysis of facial expression coding and body language measurements and pupilometrics examinations etc. [28]. Qualitative data contains information which reveals thoughts together with emotional or experiential components from individuals. The collection of such data commonly happens through text material and interviewing along with direct observation methods. The evaluation of emotional states through sentiment analysis would require the identification of sadness, anxiety and hopelessness within text-based data. Through qualitative interviews or surveys people can freely share how MHW issues affect them by using their own words. Qualitative scanning of social media text would reveal first-person accounts about how individuals or specific groups experience MHW problems.

Recent studies emphasize the important role of neuroscience in providing insights into basic mechanisms of neural function [29]. Nevertheless, until now, these insights have not resulted in improved ability to engender superior assessments, diagnoses, prognoses, or treatments in MHW related conditions [13], [26], [29], [30]. Furthermore, they highlight the limited impact of neuroscience-based approaches for psychiatry on pragmatic clinical decision-making. Nevertheless, AI can supplement and complement psychiatric issues [[31]]. Inquisitive readers can refer to [[31]] for a detailed discussion on the role AI/ML as well as analytics of big data, behavioral data, real-time data, etc. in MHW services. Indeed, such a fusion of data and technologies form the foundations of the fourth industrial age [[32]].

Recently, the research community has shown significant interest in employing multimodal approach to such MHW issues. Some of the recent studies would include: [33] Uses two types of data to screen a speaker for depression, it uses deep learning techniques where CNN is used for voice and transformer is used for text. [34] fuses text, images and acoustics data to detect the bipolar disorder and depression in patients. Meshram & Rambola uses the psychiatric illness history, textual descriptions and images to detect depression scale in patients [[35]]. Fazeli et al. extended the multimodal scope of data collection, using a smartwatch during an exercise session, with heart-rate readings, number of floors climbed, BMR kilocalories, distance traveled, activity levels, aggregated HRV measures, SPO2, respiration rate, and HRV-based readings [[36]]. The data collected is using a smartwatch. Bucur et al. use a different route to use social networking data uploaded by an individual to diagnose depression with data comprising of text and images [37]. Thati et al. use mobile usage data which includes Acceleration, Application Usage, Brightness , Bluetooth encounters, Gyroscope values, GPS/ location and Screen unlocks [38].

2.3. Significance of Triangulation in MHW

The research technique of triangulation uses numerous sources and approaches to investigate phenomena [39], [40], [23], [41]. Researchers aim to improve both finding validity and reliability through cross-checks of different information sources [39]. Solid triangulation design requires integrating multiple research techniques or data types since this approach helps researchers reduce single-source limitations and biases. Multiple MHW research environments would benefit from full triangulation practice which includes using both quantitative and qualitative data collection with mixed-method methodologies. The field of mental health research demonstrates an increasing acceptance of using triangulation because it combines various complementary data sources and research methods which lead to more precise understanding of complicated mental health

issues [42], [43], [44], [45], [46], [47]. The methodology for social and MHW research receives extensive theoretical and methodological support from Denzin (2012) [43].

Triangulation itself could be multi-faceted. Data triangulation employs various research instruments including interviews as well as observations and documents and surveys to achieve thorough understanding of research topics according to [43], [48]. Through different data collection methods researchers can identify both matching findings and points of difference [43]. Methodological triangulation combines multiple research methods and approaches to analyze the same research question as explained in [42], [43]. Research studies implement qualitative together with quantitative methods to obtain exhaustive and integrated data analysis results [43]; [45]. Various researchers or investigators collaborate independently under the framework of investigator triangulation to gather and interpret data [43]. The utilization of multiple perspectives reduces researcher biases while preparing an unbiased interpretation of research [43]; [45]; [47].

Other than clarity purposes we must stress that the concept of triangulation serves to verify findings using multiple sources or methods, yet the multimodal approach handles the analysis of different data modes together. Triangulation serves as a broader concept which includes the multimodal approach as one of its fundamental components.

2.4. AI in MHW

AI demonstrates impressive potential for clearing barriers which enable MHW to reach positive accessibility standards as well as efficiency and efficacy goals [13], [15], [30], [49]. Integrated artificial intelligence in MHW activities can enable new discoveries of support solutions to help societal needs [50, 51]. Various recent articles provide good reviews on benefits and challenges related to the use of AI in healthcare including MHW at both individual and societal levels [30, 31, 52]. Furthermore, good reviews to acquaint the reader with key terms related to AI and psychiatry are available [53]. AI has been used, and prescribed for use, in prevention, risk stratification, early detection, prognosis, diagnosis, management, and rehabilitation related to MHW [54].

This section focuses on literature related to the use of AI in diagnosis, prognosis, and treatment of MHW issues. Importantly, all these studies concur on the efficacy of employing AI in MHW with a careful consideration given to bias, privacy, transparency, ethical concerns and validation of the techniques. In this context, Eco-SMART is a framework that would furnish comprehensive and valid delivery of mental health care to all users, contributors, and other stakeholders of MHW services. It offers a seamless, undisruptive, and controlled integration of the AI-based services in MHW applications.

2.4.1. AI in MHW Diagnosis

AI offers significant potential to play a substantial role in the diagnosis of mental health conditions in individuals and societies [55]; [56]; [17]; [57]. Consequently, the use of AI in identification and determination of MWH related conditions has been increasing [55]. Abd-alrazaq et al. (2022) provides a systematic review of such literature [55]. Table 1 shows some recent references in this regard. For instance, some researchers use disease-centric knowledge-graphs facilitate exploring relationships among various knowledge resources and effectively answering clinical queries [[58, 59]]. Kakde (2023) uses fMRI data with parcellation and a logistic regression to classify the patients for schizophrenia [[60]].

Priya et al. (2020) presents a comprehensive study of the utility of machine learning algorithms with multimodal data to estimate the transition to psychosis from an on-set depression condition [61]. It uses a number of machine learning algorithms with clinical data, neurocognitive, neuroimaging, and genetic information. The article by Yu et al. (2024) presents an organized study showing how NLP and LLM systems used in medical diagnosis alongside patient engagement and electronic health record management and personalized medicine lead to improved outcomes for healthcare [54].

Different AI methods utilized for MHW diagnosis consist of Machine Learning (ML) Systems [62]; [63]; [64]; [65]; [56], Fuzzy Logic (FL) [66]; [64]; [65]; [56], Evolutionary Search [56, 64, 65], Natural Language Programming (NLP) [67]; [54], Large Language Models (LLM) [54], Generative AI (GenAI) [68], Decision Support Systems (DSS) [64]; [65]; [56]. The current usage of artificial intelligence alongside machine learning in MHW application diagnoses shows limitations according to studies conducted during the last few years [27, 69, 70].

2.4.2. AI in MHW Prognosis

Prognosis relates to the prediction of the likely course and outcome of a medical issue or ailment. In MHW, prognosis is deemed a more challenging task due to the need for looking quite far into the future while considering numerous dynamic factors that may or may not be controllable [[71]]. Nevertheless, AI also offers significant potential in the prognosis of mental health conditions in individuals and societies. Consequently, AI has been used in the prognosis of MWH related conditions. Table 1 shows some recent references in this regard.

2.4.3. AI in MHW Treatment

AI also offers significant promise to play a substantial role in the treatment of mental health conditions in individuals and societies. Consequently, AI has been used in overall care and treatment of MWH related conditions. Table 1 shows some recent references in this regard.

2.4.4. AI-Based Conversational Agents in MHW

Conversational Artificial Intelligence, particularly AI-based conversational agents (CAs), have also been gaining traction in MHW. Li et al. (2023) provide a systematic review and meta-analysis of their impact on the effectiveness of AI-based CAs in improving mental health and factors influencing their effectiveness and user experience [[72]]. This meta-analysis of literature indicates that the use of AI-based CAs results in a significant reduction in symptoms of depression [[72]].

Digital mindfulness-based interventions using tools like Chatbots have been highlighted as promising approaches to improve mental health outcomes like stress (eg, [[73]; [74]; [75]]), mindfulness (eg, [[73]; [74]; [76]], subjective well-being (eg, [[74]; [76]]), as well as in emphasizing the role of emotion regulation (e.g., [[77]]).

Table 1. AI in MHW diagnosis, prognosis, and treatment: A Summary of References.

Diagnosis	[70], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [82], [72], [89], [90], [91], [92], [51], [57], [93], [94], [95]
Prognosis	[63], [71], [71], [96], [91], [97], [98], [99], [51], [57]
Treatment	[63], [71], [96], [91], [97], [98], [99], [51], [57]

2.5. Existing MHW Apps/Products and Limitations:

There is a significant growth trend in the number of MHW related cloud or smartphone apps as discussed in the previous sections. Presently, over 10,000 MHW apps

are available online through the Apple App Store, Huawei application store and Google Play Store [20, 100, 101]. In addition, projections show that the global digital mental health market will grow from USD 2.6 billion in 2021 to USD 19 billion by 2030, projecting a compound annual growth rate of 21.1% [102]. This exponential growth is not only indicative of the potential of the market but also the growing demand for digital mental health solutions. The increased global awareness about mental health, technological adoption, and the shift in healthcare approach from traditional methods to patient-centric digital solutions are the factors leading this growth in MHW apps [20].

Existing MHW related apps have some considerable shortcomings, impeding the wider acceptance and proliferation of such apps, as pointed by various researchers. Lecomte et al. give a review of the reviews of mental health applications and suggests that most of the platforms are not clinically tested [[103]]. O'Loughlin et al. review transparency in data privacy and security of the mobile application e-mental health, particularly depression [[104]]. The findings are disconcerting as the e-mental health mobile applications do not focus on the privacy data in case of identifiable and unidentifiable data collection. Furthermore, there are challenges related to the platform like, scalability, adaptability, passivity, searchability, etc. These shortcomings frequently result from the absence of a unified, intelligent, scalable, and adaptable platform that could simplify the process of connecting an app with various services. Provision of such a platform would enable MHW app developers to focus on building the core functionality rather than dealing with complex backend infrastructure. Eco-SMART will fill such gaps.

Interestingly, there is a slew of FDA approved apps for medical diagnosis and treatment, including apps in the MHW domain [105]. However, such MHW apps also require some platform for effective functioning and interoperability, as also emphasized by FDA [105]. Consequently, we deem the number as well as the growth rate in MHW apps and users as worthwhile determinants in estimating the demand for platforms like Eco-SMART, which will act as a middleware between different MHW apps, facilitating communication and data exchange between different apps or systems.

2.6. Existing General Purpose Platforms

There are several platforms available for developing mobile or cloud-based apps. Seiferth et al. give the guidelines for e-mental health for researchers and practitioners. It reviews and reflects on e-mental health assessments and intervention [106]. They discuss the four digital services covered by e-mental health: information provision, assessment for screening and monitoring, intervention and social support. Moreover, Smoktunowicz et al. [107] provides a comprehensive review on the terminologies used in internet-based e-health interventions and used a Delphi method to posed ideas to standardize the nomenclature. It is important to gauge the effectiveness and harness the benefits a standard for MHW platforms is needed. Torous et al. give an architecture of the learning healthcare system that includes mental health [108]. It presents an adaptable system that collects the data, analyzes data and involves humans to device personalized care plans. Another comprehensive study evaluated the internet-based mental health application platform end-to-end [109], and it evaluates the iterative process comprised of psychological treatments, use of artificial intelligence, human-computer interaction, and software engineering. Furthermore, the study provides direction for research and development for mental health application platforms.

Although the literature talks about the general architecture of a learning healthcare system that includes mental health, there seems to be a dearth of literature discussing the architecture of a dedicated MHW app platform. Even though there are a number of viable options for developing apps, there is a dearth of platforms for supporting MHW apps. For instance, iOS and Android Platforms offer generalized support in generic app development. The Apple iOS platform offers a robust framework for building mental health applications. The platform gives developers access to tools such as HealthKit [100]combined with ResearchKit and CareKit [101] for adding health functions to their products along

with secure data collection mechanisms. The Android platform gives developers numerous development options alongside a wide base of users for mental health applications. Google Play Services allows developers to use its APIs to track fitness data and manage user authentication and store healthcare information. Particular platforms exist for mental health applications to help developers develop specialized features instead of creating everything new. Such platforms come with predefined components as well as data management features and connectivity capabilities. Three widely used platforms for mental health applications are MyStrength [110] for Phone and Tablet, Lantern for Phone and iPad Tablet [111] and Headspace [112]

MHW app development and deployment services are accessible through cloud platforms including Amazon Web Services (AWS) [113] , Microsoft Azure [114] and Google Cloud Platform (GCP) [115] along with their respective competitors. These cloud platforms give developers access to scalable servers with data storage capacities and analytics tools as well as machine learning functionality needed to create scalable applications. The selection of your platform should start with target audience analysis along with necessary feature assessment and scalability projections and data security needs with development resources currently accessible. Relevant regulatory guidelines require compliance with strict regulations and users need protection for their sensitive data.

3. The Eco-SMART Platform

Eco-SMART is a comprehensive end-to-end system designed to permit plugging in MHW apps to an eco-system of such apps in a seamless manner. It consists of a scalable and extensible platform with service-oriented architecture (SOA). SOA is a design approach that emphasizes communication between different software components or services to achieve a loosely coupled and interoperable system.

Heterogeneous data collection starts by gathering text along with images and audio files along with physiological data and contextual information. Advanced analysis methodologies function in real-time while processing these collected data through techniques such as sentiment and emotion analysis as well as behavioral pattern detection and predictive analytics. Healthcare specialists together with users obtain descriptive mental state information through visualizations presented by the system because it continuously monitors subjects. A comprehensive approach for early MHW detection enables healthcare personnel to develop well-timed interventions which simultaneously improve mental health education for all individuals.

The platform brings together multiple data streams with analytic methods to provide live behavioral monitoring and predictive alert systems that enhance awareness and fast MHW diagnosis. This platform provides users and healthcare practitioners with useful data which results in maximizing mental health results while using a data-driven approach to track mental status. The eco-system of an MHW app operating on the Eco-SMART platform can be observed in Figure 1 as an abstract representation.

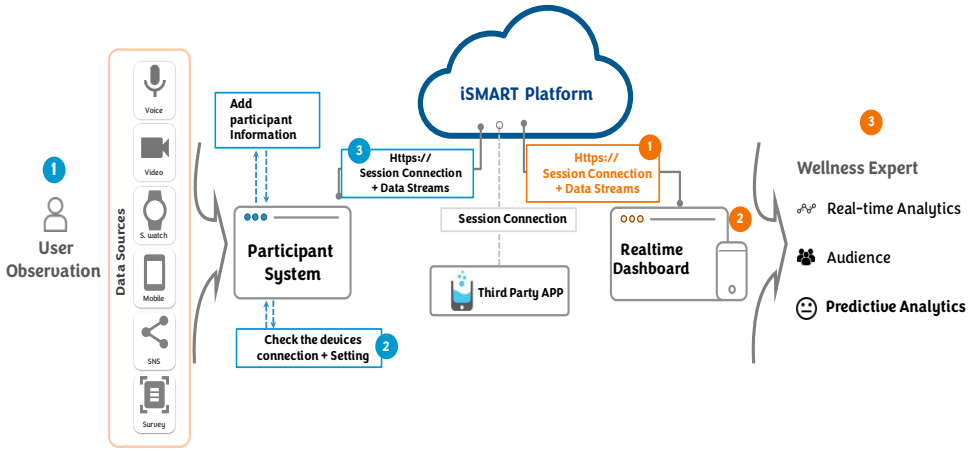

Figure 1. An Abstract view of an MHW App running on the Eco-SMART platform

Figure 2 shows the detailed system architecture of the proposed platform. In the following, we provide detailed descriptions of various components of Eco-SMART platform layers in detail along with the respective objectives and workings in detection of MHW issues. We will employ MDD detection and support as a demonstration case study to frame descriptions of all layers and modules within a realistic, identifiable, and understandable context.

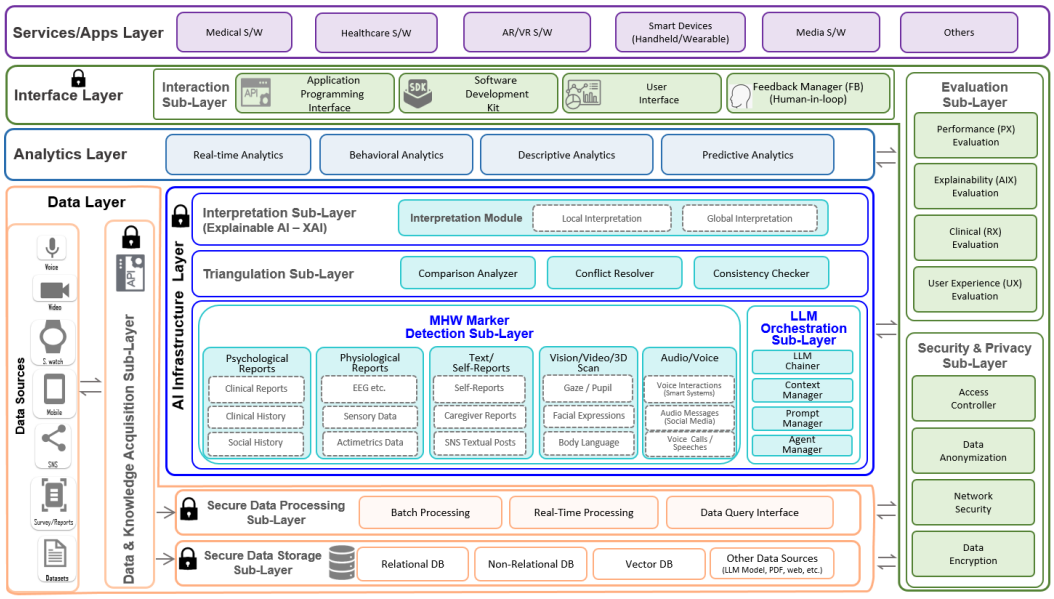

Figure 2. The detailed architecture of the proposed Eco-SMART platform.

3.1. Data Layer

The Data Layer sets its target to build an infrastructure system which collects and processes and stores and analyzes multiple data forms to recognize depression symptoms. This system depends on multiple data sources to extract valuable information through its pre-trained models. Data layer investigates deep into the particulars continuously collecting raw sensory data from various multimodal data sources in real-time, ensuring device independence, enriching the context by incorporating a wide range of sensory devices, and providing persistence to handle the substantial volume of heterogeneous and multimodal data associated with user sessions. To achieve the goal of data layer, we need to consider multiple aspects such as continuously sensing, acquiring, enriching, and persistently storing multimodal raw sensory data, while maintaining the device independence and the user data privacy.

Users can interact with the system through a query interface to obtain relevant information and insights for depression detection and research purposes. Whether users are keen on understanding the interface between different devices or the fusion of multimodal data to gain deeper insights or even the application of AI in detection of MHW issues using multiple facets and multimodal data.

Here we provide a high-level breakdown of the key components and functionalities of the data layer. The Data/Knowledge Acquisition & Integration Layer stands as the fundamental component of the data layer because it connects to video cameras alongside mobile sensors as well as social networks and domain experts. The "Data Processing Layer" performs a meticulous process on the acquired data. The "Data Storage Layer" manages data storage and retrieval of processed information. The subsequent section contains an in-depth analysis of these three sub-layers.

3.1.1. Data & Knowledge Acquisition and Integration Sub-Layer

The data layer interfaces with a wide range of sensory devices and data sources. These sources may include, but are not limited to: video cameras (image & voice) and feeds; Smartwatches and wearables devices; Mobile phones and their sensors (accelerometers, GPS, etc.); Social Networking Services (SNS) APIs for collecting data such as posts, comments, and messages; Surveys or Questionnaires administered to users; knowledge from domain expert; caregiver and self-reports, etc.

3.1.2. Data Processing Sub-Layer

The sub-layer requires data preprocessing and transformation to perform quality control and normalization as well as extract relevant features. The selection of preprocessing techniques depends on the data source while its format determines how techniques should be applied. The data arrives in both batch and streaming patterns. The Figure 3 diagram illustrates that batch processing manages data in big quantities at once yet streaming processing handles in-coming data streaming as well as near-real-time evaluation.

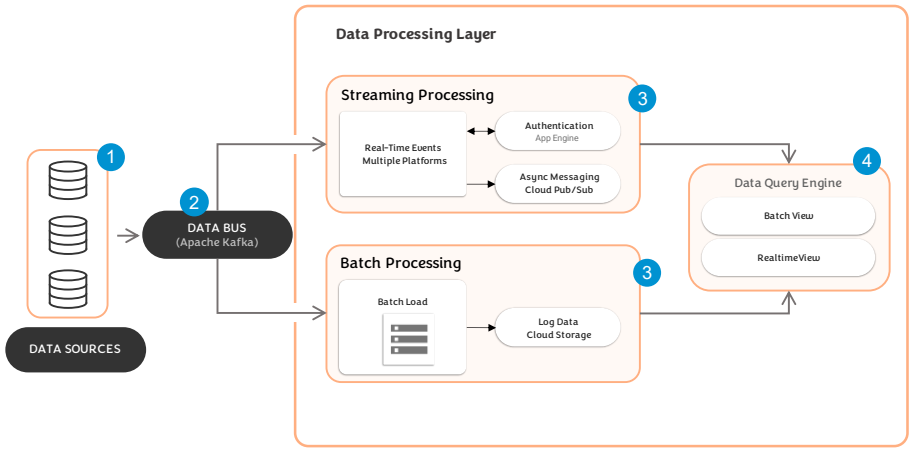

Figure 3. Data Processing Layer.

The data collection method under Batch Processing accumulates extensive data sets for subsequent processing after data collection duration. As a part of the process, you clean data then filter and transform data while utilizing parallel computing to manage large data sets across multiple nodes or processors. Batch processing is beneficial in feature selection, especially when dealing with large multimodal datasets.

Real-Time Processing handles real-time streaming processing of events as they happen, and the data are ingested in small but continuous streams. Streaming processing effectively handles timestamp data, where events within a stream are associated with

timestamps. The technology of streaming data enables operators to manage time in windows and enables the processing of events through their defined constraints.

Through its Data Query Engine interface users along with other system components gain access to processed data by using an API. The interface enables users to retrieve all types of data from historical records or real-time events or defined time segments.

3.1.3. Data Storage Sub-Layer

Data storage sub-layer functions as an essential fundamental component of the platform because it manages data storage and retrieval operations. The platform effectively deals with extensive data quantities while expanding to meet growing user requirements. The data storage sub-layer takes responsibility for retaining processed information within relational databases and Non-relational data-bases and vector databases. The layer also functions as a storage system that maintains pre-trained models across different phases of MHW identification in model-bases.

3.2. AI Infrastructure Layer

The AI infrastructure layer supports explainable, real-time, and privacy-preserving automation for MHW applications. It seamlessly operates with DL and dynamically employs state-of-the-art AI and machine learning algorithms for analysis. It is composed of loosely coupled modules of MHW marker detection, model orchestration, and interpretation. In general, this layer consists of various sub-layers like MHW marker detection, model orchestration, triangulation, and interpretation, as described in the following discussions.

3.2.1. MHW Marker Detection Sub-Layer

The sub-layer functions as a crucial component to optimize the usage of multiple Large Language Model (LLM) instances thereby improving scalability and performance across various applications.

Several structured self-reports combined with clinical assessment reports which include psychological and physiological reports help enhance MDD detection and utilization. The combined method enables doctors to make precise diagnoses while designing individual treatment strategies along with monitoring MDD therapy outcomes effectively. In addition, the analysis of both structured and unstructured messages posted by the subject on various social media could be used in extraction of meaningful insights and detect MHW markers from the subject-generated text-based and image-based contents.

Similarly, the use of video/vision-based data allow detection of MHW markers via analysis of facial expressions, gaze analysis, body language interpretations, etc. The evaluation of facial expressions together with gaze measurement enables researchers to obtain immediate knowledge about the emotional state of subjects. This component tracks and investigates minimal alterations in facial expressions together with gaze movements for precise mental wellness evaluations of individuals. Body language interpretations deliver important information about a subject's emotional condition to researchers. The detection of tiny signals from body language helps substantially in understanding both emotional states and physical wellbeing of the subject.

Detection of MHW markers through audio signals would transform into a vital system by examining changes in human voice patterns. A system would analyze user voice patterns to determine emotional states for improving mental well-being assessments. Under the model training process machine-learning models learn audio data through training activities. The trained model establishes the base for recognizing emotions through voice-based algorithms which detect emotions such as anger and happiness and neutral and sadness. Non-Speech Reduction technologies enable the detection of spoken communication from silent or noisy conditions. Audio recordings serve as extensive MHW data sources when they contain recordings of human conversations together with speeches

presentation content from social media messages and customer service calls and meetings and interviews and voice commands for smart systems.

3.2.2 LLM Orchestration Sub-Layer

The sub-layer acts as a fundamental feature to manage and use numerous LLM instances for achieving scalability with reliable performance across various application domains and knowledge sources. In MDD detection the LLM Orchestration sub-layer performs all necessary tasks to deploy and operate various models in combination for effective execution. Through LLM Orchestration, users would have access to multiple LLM models which include both open-source and commercial types. Users would gain the power of selecting appropriate models among their available options. The system center would contain the LLM Models which integrate cutting-edge language models alongside AI technologies. The system would unite modern language models with integrated capabilities to operate along with LLM Chainer and Context Manager and Prompt Manager and Agent Manager modules.

In an agentic AI framework, the LLM Orchestration Sub-Layer functions as a central intelligence that oversees multiple specialized AI agents—each powered by advanced language models. This approach ensures scalability and consistent performance across various domains, including sensitive scenarios such as Major Depressive Disorder (MDD) detection. By allowing users to choose from diverse open-source and commercial LLMs, the system adapts to both rapid prototyping needs and intricate, domain-focused tasks.

At the core is a System Center, which houses the AI agents and their respective models. The LLM Chainer Agent links outputs from one LLM to the next, enabling sequential operations: for instance, generating a concise summary of a lengthy clinical report, then using that summary to generate relevant queries. Simultaneously, the Context-cum-Memory Manager Agent autonomously handles data storage and retrieval, preventing redundant processes (such as repeated translations) and preserving dialogue continuity. The Prompt Manager Agent guides user interactions with the LLMs, recommending prompt structures that elicit precise or domain-specific responses.

Finally, the Agent Manager Agent enforces constraints and logical standards across all AI operations, ensuring the system's output remains accurate, stylistically consistent, and ethically compliant.

Together, these AI agents work under the Orchestration Sub-Layer's directive to provide a cohesive solution that is context-aware, adaptable, and robust. In MDD detection, for example, this agentic ecosystem can integrate patient data, relevant literature, and ongoing clinical interactions to generate comprehensive insights without compromising on reliability or domain-specific integrity. Such a framework exemplifies how advanced language models, when organized into autonomous yet collaborative agents, can transform raw computational power into sophisticated, reliable outcomes for a wide range of applications.

3.2.3. Triangulation Sub-Layer

Our approach would employ an AI-Augmented Triangulation for detecting MHW issues. In this context, the term triangulation refers to the use of multiple methods and/or data sources in qualitative research to develop a comprehensive understanding of the phenomena. Triangulation layer would comprise of components that permit employment of multiple sources of data or multiple methods to validate a specific MHW issue, as shown in Figure 4.

Several types of data analysis combined together increase depression detection accuracy through enhanced reliability of results. Different types of data about stress levels and emotional expressions combine to highlight their mutual importance in clinical assessment which leads to improved intervention outcomes.

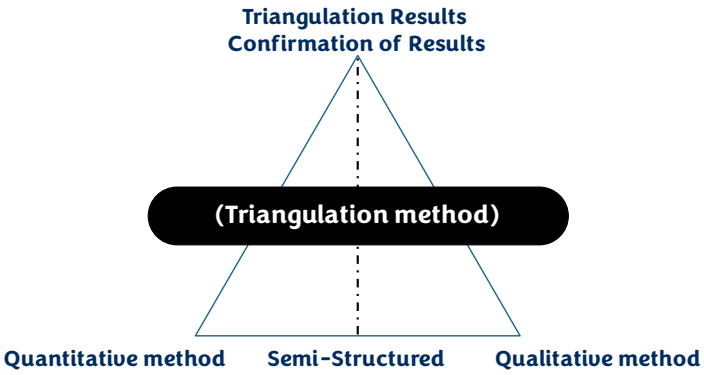

Figure 4. Triangulation Process.

3.2.4. Interpretation Sub-Layer

An AI interpretation sub-layer facilitates the explanation of the output of AI models. This critical sub-layer may also be known as an inference layer, explainability layer, or decision-making layer. The fundamental purpose include helping users make sense of the predictions or decisions made by the AI models as well as enabling the system to take appropriate actions, bridging the gap between raw AI outputs and meaningful actions or results.

The interpretation layer accepts output from the AI model through predictions as well as classifications and recommendations alongside relevant information types. The input integrates probabilities and confidence scores and raw data originating from AI model output. The interpretation layer operates on AI model outputs through established rules and thresholds together with domain-based knowledge for interpretation purposes. The interpretation stage incorporates a combination of application context together with user preferences and industry guidelines and best practices and regulatory requirements to formulate its decisions. Supplementary post-processing methods would operate through the interpretation layer to enhance and modify the results generated from AI models. The process would entail merging predictions with techniques to refine results while removing anomalous datapoints alongside making decisions for handling rare situations to maintain alignment between targets and limitations.

The interpretation layer would assess the confidence or reliability of the outputs of AI models. The provision of reliability scores, confidence intervals, or uncertainty estimations would indicate the level of trustworthiness associated with the output, which are very useful to decision-makers and users.

The interpretation layer produces suitable actions or output which result from analyzing output together with its connected decisions. Through the interpretation layer the system executes multiple functions which include starting business operations as well as showing output results and displaying warnings and interacting with other AI-system components. The interpretation layer provides support for feedback mechanisms that allow users and systems to record their feedback and actual outcomes regarding *actions taken*. The feedback system helps improve AI models while allowing the interpretation process to adjust itself throughout time. The feedback loop provides dual benefits which improves decision-making functions of the interpretation layer while strengthening overall system performance.

The decision-making process requires simultaneous evaluation of local and global interpretations in this setting. When using local interpretation methods the system analyzes why a model makes certain predictions about a single input observation. The input features which influence model decisions receive emphasis through utilization of feature importance analysis combined with saliency mapping techniques or attention mechanism methods. Local interpretation delivers detailed accountability regarding individual

model cases yet it lacks the ability to detect broader trends and patterns. Global interpretation delivers a comprehensive understanding of how a model processes information through every data point in its collection. General patterns as well as relationships and trends in predictions help achieve this interpretation. Global interpretation usually relies on methods such as feature importance analysis together with performance measurements and decision boundary visualization and internal representation analysis. Global interpretation provides researchers with overall knowledge about the predictive model throughout the dataset while pointing out potential weaknesses and best features.

3.3 Analytics Layer

The Analytics Layer serves as the core intelligence hub in Eco-SMART, encompassing a range of advanced data analytics and visualization techniques and algorithms designed to extract meaningful information from multi-modal data processed data. This layer operates in real-time, ensuring that insights are timely, actionable, up-to-date, and responsive to evolving mental states of users aimed at early detection and support in MHW issues. We will provide a variety of analytics including Real-time Analytics, Behavioral Analytics, Descriptive Analytics, Predictive Analytics, etc. In the following, we provide some perfunctory information on these pertinent analytics.

3.3.1. Real-Time Analytics

The Real-Time Analytics module evaluates multisource and multimodal data simultaneously to deliver instant user psychological and behavioral state information. The application of advanced analytics allows immediate mental health support in addition to intervention services for people experiencing MHW symptoms. Real-time Analytics module allows the fulfillment of different essential targets. The tool detects present behavioral patterns by tracking changes between communication patterns along with facial expressions as well as physiological measurements alongside voice tonal shifts since these indicators can show symptoms of mental illness. The system should provide automatic alert and notification features after identifying major emotional distress indicators to support instant help. The system offers real-time visibility of data-derived knowledge through intuitive dashboard systems that both users and medical practitioners can easily interpret to support immediate decision processes. The approach evolves analytics by analyzing incoming information to develop better accuracy and efficiency performance.

3.3.2. Behavioral Analytics

The Behavioral Analytics module aims to study and comprehend users consisting of patients along with caregivers and healthcare staff. Through understanding the specific actions along with choice systems and anticipated reactions and essential requirements of multiple stakeholders this module helps develop better implementations for Eco-SMART.

The module supports multiple essential goals. The evaluation of how patients and caregivers together with healthcare professionals engage with Eco-SMART when they use the app is the first priority of this analysis. The platform monitors users through their profile paths and monitors their content usage while tracking their degree of engagement. User-related understanding comes from user feedback integration along with survey results and analysis of content produced by users. Feature and content development benefits through the knowledge of user expectations. The user audience divides itself into specific groups according to their demographic traits with their behavior patterns and personal requirements. Through segmentation it becomes possible to provide customized materials and assistance. The system utilizes surveys and reviews together with direct communication tools to obtain user feedback which generates ongoing opportunities for platform improvement. The platform adopts customized features together with content which aligns with users' preferences and needs to develop its MHW support system with a user-centered approach. Platform engagement along with user retention increases when platform features specifically cater to individual user groups according to their needs and motivations.

3.3.3. Descriptive Analytics

The Descriptive Analytics module analyzes historical and real-time data to generate complete user understanding of emotional states, behaviors, engagements and MHW trend patterns derived from historical and present states. Decision-makers are better equipped and early-risk detection becomes achievable as well as MHW challenge awareness improves through the analysis of past and present user situations. Various essential objective fulfillments are supported through this module. The evaluation of past user interactions depends on multi-modal data analysis including user-generated text and images as well as platform interactions and physiological data together with audio content. Users as well as healthcare professionals can monitor emotional trends because this system enables them to track changes over time which reveals their well-being status with possible emotional triggers. The analysis evaluates user interaction with mental health content including most sought-after subject matters.

3.3.4. Predictive Analytics

Through the Predictive Analytics module healthcare organizations can forecast and predict MHW trends by combining historical data with user behavioral and multiple data sources. The module would help fulfill diverse essential purposes. The module has the capability to foresee upcoming MHW difficulties before they occur so organizations can begin preventive support measures before struggles begin. ML algorithms and predictive models can detect small behavioral shifts and sentiment variations and physiological responses which assist in predicting future MHW-related difficulties. Through intensive analysis of past user data the system would show recurring risk factors together with behavioral patterns and emotional trends and multi-modal information. The implementation of predictive models should serve to calculate the MHW risk potential of individual users. The predictive system must include an early warning mechanism that produces alerts anytime the risk assessment points toward an MHW issue so that intervention efforts can occur swiftly. Sixth, visualization of predictive insights in a presentable and interpretable format.

3.4. Interface Layer

Users connect with systems through a system interface which demonstrates the complex interaction between the two. The Interface Layer establishes the main point of user interaction by determining how effortlessly users can navigate and interact with system behaviors. The Interface Layer of the provided system diagram includes three sequential sub-layers composed of Interaction Sub-Layer and Evaluation Sub-Layer and Security & Privacy Sub-Layer. The following part focuses on detailed explanations about these sub-layers starting with their operational characteristics and important aspects.

3.4.1. Interaction Sub-Layer

The Interaction Sub-layer includes four essential parts namely Application Programming Interface (API) together with Software Development Kit (SDK) and User Interface (UI) and Feedback Manager (FB). The modules operate in union to generate the required interactivity for developers and users.

API functions as an interface set of protocols and tools enabling various software components to exchange communication. Through its bridging mechanism API facilitates different systems to interact without forcing them to understand the underlying system operations. The key objectives of the API setup include several points. Through its "Gateway to Interaction" function API allows both developers and systems to access platform capabilities. The protocols established by APIs permit standardized data exchange no matter what structure external systems adopt. Weak API develops "Extension & Integration" capabilities that enable developers to create extensions or third-party applications

which interact seamlessly with Eco-SMART features. The platform develops a environment that lets developers build applications which meet particular requirements. The controlled access through APIs lets external entities query modify and manipulate data according to their needs ensuring dynamic data access with real-time responses for "Data Accessibility & Manipulation".

SDK functions as a set of development tools together with programming software that enables developers to build applications for particular system frameworks through specific platforms. The main objectives of the SDK Component include targeting specific goals. With its name of "Toolbox for Development" SDKs deliver a complete set of development resources that include libraries together with code samples and extensive documentation for developers. These development tools both speed up development work and enforce proper programming standards in all applications. SDK development restricts itself to the Eco-SMART platform hence MHW applications built through the SDK can perform optimally on this platform. All MHW apps developed for the Eco-SMART platform maintain consistency through the standardization of SDK tools and guidelines which guarantees homogenous appearance and user interactions with enhanced user comprehension.

The User Interface (UI) defines the area where system users interact with the system through all visual and audio-based components. A user interface emerges from all visual and audiovisual and touch-based components which users interact with. The system presentation interface determines user perception and system interaction through its visual design elements. Users achieve easy organization and understanding of system elements through a properly designed UI. A dynamic user interface provides active feedback that shows users both feedback on current user interface modes and their interaction consequences to create feelings of system control. The UI serves as an essential factor for user engagement together with system retention levels. System success often depends on a design which offers both usability and aesthetic attractiveness and user understanding.

The system uses the Feedback Manager (FB) to merge human inputs into its operational processes. The controller enables human involvement during system operations thus combining human oversight with system validation and intervention capabilities. The system achieves better interaction quality through time as it integrates human feedback which helps identify errors and adapt to different situations. Automated systems lack human intelligence which helps make better decisions because people bring vital information about situations across the board. A key component of learning systems that involve ML or AI requires feedback from human operators to improve system training. System refinements which result from human efforts to validate system decisions or provide corrections produce improved algorithm accuracy for future predictions.

In addition, FB facilitates active learning of the AI models. The iterative nature along with adaptability in machine learning makes active learning a method where the model works with users and other sources to get desired output to identify important data points. The human supervision during "hu-man-in-the-loop" operations enables excellent active learning by refining both LLMs and other Deep Learning (DL) models. Active learning serves to connect the automated capabilities of machines with human-based knowledge.

Active learning requires data labeling because models during their early development identify uncertain data points to request human labeling of those points from humans. The iterative feedback system enables the model to process limited labeled examples more efficiently while conserving time and cost. User trust should increase when they become aware that human oversight exists for essential system processes because this ensures minimal automated decision risks through human judgment.

3.4.2. Evaluation Sub-Layer

The Evaluation Layer serves as a comprehensive assessment module, meticulously analyzing and gauging the performance, explainability, clinical relevance, and user experience of implemented systems, ensuring optimal functionality and user-centric outcomes. The multifold goal of the Evaluation Layer includes Performance Evaluation (PX), Exaplainability Evaluation (AIX), Clinical Evaluation (RX), User Experience Evaluation (UX), Feedback Integration, Alignment, etc.

The *Performance Evaluation* module assesses how well the AI system is performing in terms of accuracy, speed, and other performance metrics. It may involve techniques such as cross-validation, precision-recall analysis, and ROC curves. The *Explainability Evaluation* module evaluates the transparency and understandability of decisions by the AI models [[116]; [117]; [118]; [119]; [120]]. Explainability reduces the otherwise inherent "black-box" nature of AI models by enabling interpretations by humans. The *Clinical Evaluation* module ensures the system outputs are clinically valid by assessing if conclusions of AI align with medical knowledge, standards, and practices. The *User Experience (UX) Evaluation* module focuses on the end-user and evaluates intuitiveness and user-friendliness of the system interface, which are key determinants of user satisfaction with the system interactions. The Feedback Integration module creates a loop which integrates feedback from various stakeholders into the system for continuous refinement and improvement. The Alignment with Objectives module ensures that the system operations and outcomes consistently align with the overarching objectives and goals of the organization or initiative it supports.

3.4.3. Security & Privacy Sub-Layer

The service layer implements security policies and controls both authentication processes of users and authorization protocols for clients. The system executes specific operations only after validating user rights and permission credentials. MHW apps handle user-sensitive data so security protocols and privacy rules and research ethics present essential requirements for this data.

This notion also applies to Eco-SMART platforms, which handle sensitive and personal information, where any breach or unauthorized access can lead not only to financial or reputational repercussions but also to severe ethical concerns and potential harm to users. O'Loughlin, et al. (2019), review transparency in data privacy and security of the MHW related apps [[104]]. However, research suggests that in case of MHW apps, privacy and transparency are not given due consideration, resulting in lack of trust and confidence in the mind of users. Consequently, the security should be inherently built into the system design and architecture, the underlying platform, as well as various add-on modules.

Protecting patient confidentiality and trust is a top priority. Consequently, the core objective of Security and Privacy Layer is to establish stringent protection measures for data storage and transmission. This layer consists of multiple sub-components including Access Controller, Data Anonymizer, Network Security Module, and Data Encryption Module. This not only ensures availability of this data only to authorized personnel but also compliance to relevant medical and health privacy standards and regulations.

The *Access Controller* operates as the primary defense against unauthorized access, serving as the digital equivalent of a security guard at the entry point. It would involve User Authentication Methods for verifying the identity of a user before granting system access using password authentication, biometrics, multi-factor authentication etc. A Role-Based Access Control (RBAC) would restrict system access based on roles. The resultant Permissions and Privileges would define the operations that an authenticated user is authorized to perform.

The *Data Anonymizer* ensures user data confidentiality by transforming or encrypting it in such a manner that personal identifiers are eliminated or altered. This involves Tokenization process in which sensitive data is replaced with non-sensitive placeholders or tokens. The actual data is stored securely elsewhere so that data is not directly accessible.

Data Masking conceals original data with modified content while retaining the data format. For instance, a social security number might appear as XXX-XX-1234. Generalization process will reduce the granularity of data, such as changing exact ages to age ranges.

The *Network Security Module* encompasses measures to protect the system's integrity from external and internal threats. Firewalls function in this module to establish security barriers which separate trusted local networks from untrusted external networks. Content-based security policies determine how these firewalls block and allow internet traffic movements throughout the system. IDS systems track network traffic to discover suspicious activities then produce alerts toward detected suspicious activities. Regular Security Audits evaluate system vulnerabilities which helps to verify that all security measures remain updated and effective.

The Data Encryption Module delivers protected data confidentiality through its transformation of original data into encoded forms. The system encrypts data with symmetric and asymmetric methods that protect both stored information and data in transmission.

3.5. Services Layer

The service layer consists of abstraction which keeps business logic and application functionality together in the Eco-SMART platform. The system would organize helpful business services into defined components which different client applications or components could access. A system that determines through its core elements how data needs to be handled to meet specific requirements. This abstraction feature includes specialized calculations, algorithms and workflows which exclusively belong to the application domain. The Eco-SMART Platform separates its business operations into a service layer to produce a flexible system architecture. Easier maintenance testing and scaling benefits are provided through such design choices.

The service layer specifically oversees transactions to achieve safe execution of multiple operations. Through its operations the layer protects data consistency and maintains integrity during operations with multiple data sources. The service layer manages all exceptions as well as error conditions while simultaneously generating essential log records. An error handling and reporting system placed in a central location provides utility during debugging and system monitoring operations. Through its services layer the system completes data validation alongside data transformation to verify input from client tools matches the business requirements and data structures. The service layer acts as a mediator between the interface layer while it communicates with databases and external APIs and all other connected services. The layer contains all rules for data retrieval and modification and persistence activities. [121]

4. Case Study: Eco-SMART for MDD Detection using Audio-Based Data

To describe Eco-SMART as an MHW app platform tangibly, we use MDD as a case study to demonstrate how MHW related apps could positively benefit from an encompassing platform like Eco-SMART. We have chosen MDD because it is one of the most common MHW issues in the world. Indeed, MDD poses challenges for society and healthcare systems due to lower patients-to-doctors ratio and the consequent delays in accessing a therapist.

Traditional methods for detecting MDD, such as clinical interviews and self-report assessments, come with inherent limitations including subjectivity and a lack of comprehensive behavioral insights [122]. In contrast, emerging approaches harness the power of advanced machine learning and deep learning techniques to analyze depression using a wide range of data modalities, including facial expressions, speech patterns, gaze analysis, and physiological data [122], [79], [123]. These new approaches using passive sensor data and machine learning techniques show promise in accurately estimating the level of depression in a patient without relying on self-reports [124]. These innovative approaches

exhibit superior performance in depression analysis when compared to traditional unimodal methods [124]. Additionally, content-based ensemble methods, using electroencephalogram (EEG) and eye movements (EMs) data, for improving the accuracy of depression detection have been proposed [79]. Overall, the new approaches offer more objective and reliable methods for depression detection compared to traditional approaches.

4.1 Data Layer Support for MDD

MDD diagnosis and therapy demands access to multimodal data for analyses such as sentiment analysis, emotion recognition from speech and facial expressions, and text analysis. MDD detection input uses data from social media in text and images, videos of patients, self-reporting forms, speech and neurological data. DL use of pre-trained model for detection of meaningful features for the MDD and the interactive access to DL helps the developers to select the interesting features.

Batch Processing and streaming in DL supports the collection and ingestion of data from users' routine activities. Data preprocessing is used for noise reduction, normalization, and scaling to prepare the data for use in the analysis techniques. Feature extraction using LSTM or BERT for text data, CNNs or RNNs for audio and video data, and traditional machine learning models like decision trees or SVMs for other types of data are integral to Eco-SMART and are plugged to MDD detection.

Eco-SMART supports a logical clock for timestamping the collected data, which is key in MDD detection and diagnosis. A time frame-based synchronization method, including Complete-sync and Incomplete-sync, is used with neurological signal data to align data within defined synchronization constraints. The model-base would also include Large Language Models (LLM) for natural language processing, vision models for image and video analysis, voice models for audio data analysis. Ultimately, the system provides a "query data interface", allowing external access to the processed data, be it historical records, real-time insights, or specific analyses, such as depression detection.

4.2 AI Layer Support for MDD

A deep learning-based MDD marker detection module encompasses sub-components employing text, vision, audio, and other multimodal data-based detection methods.

For instance, the analysis of messages posted on Social Network Sites (SNS) are used in extraction of meaningful insights and detect signs of depression from text-based and image-based contents. Within the realm of Social Network Services consumers can encounter social media, messaging applications and user-made content. Analysis of user-generated content helps identify behavioral trends alongside sentiment changes and communication patterns to support the effective management of numerous LLM instances which lead to improvement in applications and use cases performance.

4.2.1 Text and Image Analysis

Figure 5 shows the main tasks of such an AI-based MDD detection module. In this example, various relevant encoding models are used for encoding of user-generated content. These encoded data are transferred to the MDD Classifier Module containing submodules for MDD classification based on various data modalities. Numerous Textual Data Interpretation techniques study SNS textual data for the detection of emotional and psychological elements. The Sentiment Analysis techniques evaluate user-created content by detecting whether posts and images show positive emotions or negative or neutral emotional expressions. Contextual Understanding techniques produce thorough knowledge of emotional expression contexts because they analyze key elements like life events and relationships and personal experiences. Construction of this module

demands the creation of algorithms that can detect MDD markers and emotional distress signals in SNS content through sadness expressions or feelings of hopelessness and sense of detachment.

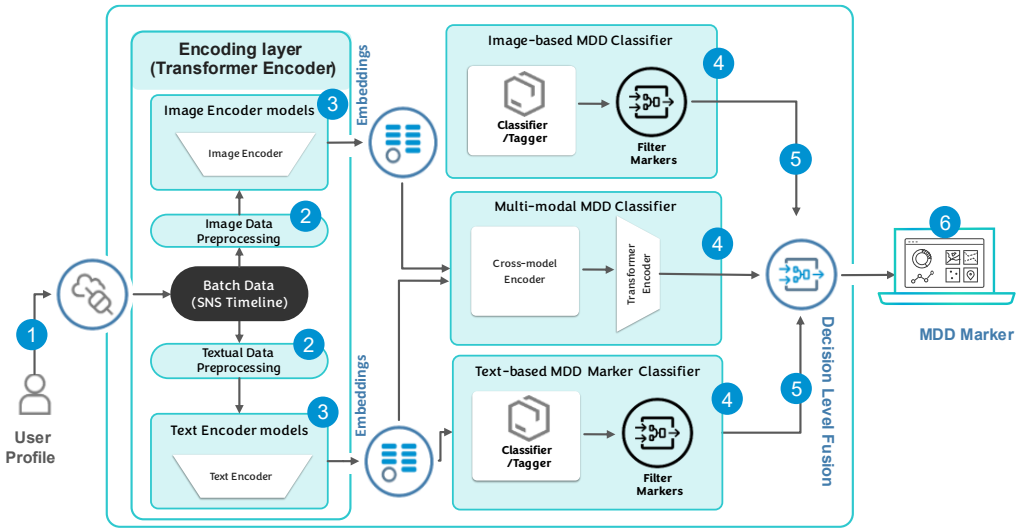

Figure 5. MHW workflow – MDD Detection Case using Analysis of SNS Posts

4.2.2 Facial Expression Analysis

Real-time emotional assessment of subjects derives valuable insights through the combination of computer vision and deep learning approaches when analyzing facial expressions. This system component strategically monitors facial expressions between sadness and happiness together with stress responses to help derive mental evaluations for persons.

The facial expression analysis component includes these essential components for successful operation. Computational algorithms and deep learning models within Facial Feature Tracking analyze facial coordinates along with expression features such as eye positions and mouth variations and brow movement. The Emotion recognition models leverage pre-trained deep learning models as an emotion recognition system that delivers precise user emotional classifications through facial expressions analysis. Real-time analysis follows an ongoing procedure which evaluates and tracks facial expressions in real time to ensure prompt reactions towards emotional state variations. The system enables connections between its components with text analysis and physical data modules to create improved comprehension of patient mental health status.

4.2.3 Body Language Analysis

The analysis combined with recognition of body movements enables researchers to understand emotional states better. The system interprets subject emotional states through analyzing body movements and gestures which leads to better understanding of emotional states along with better system interaction as presented in Figure 6. The system takes advantage of modern technological features such as sensors and depth cameras for interpreting user body expressions.

Development of an efficient system constitutes the main goals alongside creating a reliable framework to monitor mood alongside gestures and posture information. Topics range from continuously watching subject body language to identify mood and emotional experiences resulting from the subject's system interaction. In addition, extracting meaningful information from gestures of subject during its interactions with system could enrich the understanding of the emotional responses. Furthermore, analyzing postures in real-time can improve the accuracy of interpretation of emotional states.

The body language recognition component encompasses interpretation of body language, spatial structure coding for precise and efficient body part inference, pictorial structure modeling as a collection of parts arranged in a deformable configuration, and emotion recognition for registering a wide range of emotions expressed through body language.

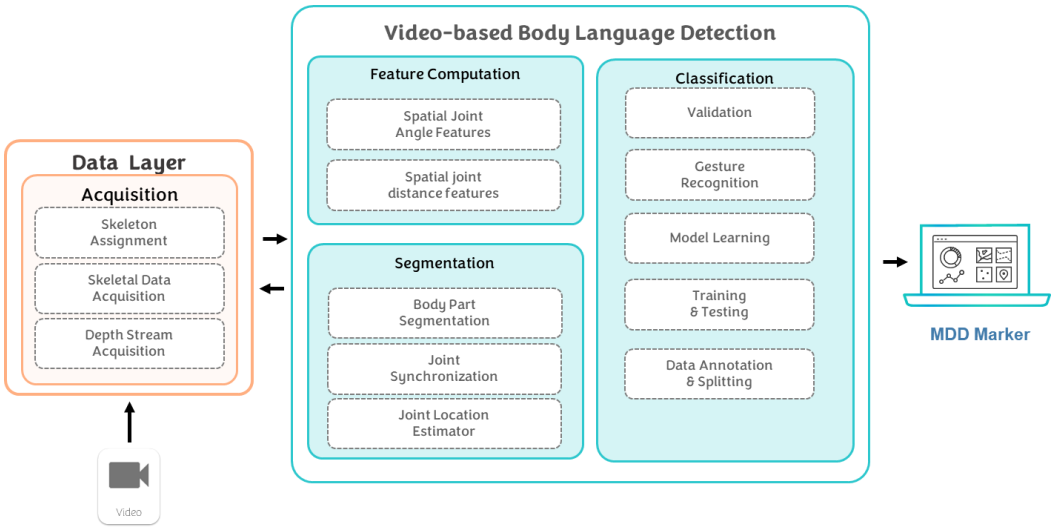

Figure 4. MHW workflow – MDD Detection Case using Body Language

4.2.4 Gaze Analysis

The analysis and recognition of gaze of subject provides valuable information by analyzing and interpreting user gaze patterns, rendering deep insights into their visual attention, interactions with interfaces, and cognitive processes. By instantaneously producing visualizations of eye tracking data and automating analysis tasks, it provides a better understanding of the mental states and behaviors of the subject and behaviors. This would contribute to the ability of early detection of MHW issues. Figure 7 shows the mechanism of incorporating gaze analysis in MHW.

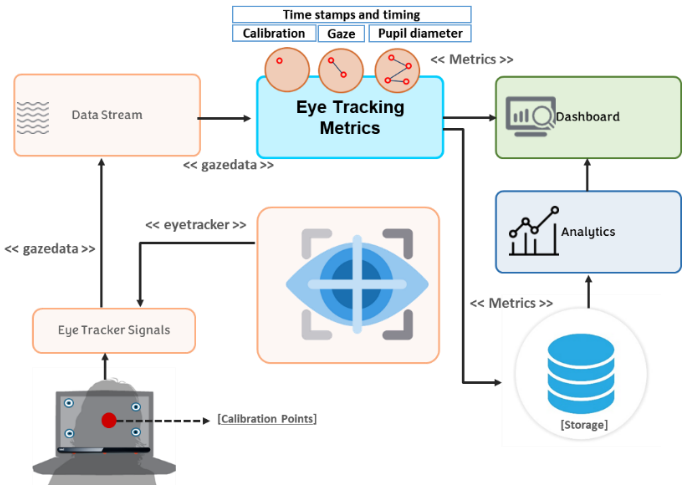

Figure 5. MHW Workflow – MDD Detection Case using Gaze Analysis.

The gaze analysis process involves multiple tasks including the following. Visualization and Automation involves instantaneously produce visualizations of eye tracking data and automate time-consuming manual analysis tasks. The gaze pattern identification involves gaze patterns of user eyes and the duration of their attention on specific content or

contexts. The analysis of interface interactions shows how users divide their vision between different interface areas which reveal their cognitive approaches toward interaction. Mapping relationships determines how eye fixation patterns combine with gaze movements and attention strength and overt attention to display hidden associations between vision attention and cognitive abilities. The focus location analysis demonstrates which parts of the subject's field of view their eyes attend and why these points land so researchers can analyze specific viewing areas. This analysis helps detect behavioral and emotional distress patterns which appear when users' attention changes by monitoring their performance behavior.

The gaze analysis process encompasses the following key elements. The Visual Attention Mechanism evaluates how visual attention functions as an information processing system through selective data recognition and unwanted data filtering. This analysis explores the core operational elements of visual attention which consist of region selection and feature dimension selection and information flow supervision and temporal transition analysis. The Visual Attention Categorization explains that visual attention operates either overtly or covertly while directing itself to object regions or object features or complete objects. Categorization of visual attention into three primary types: Selective Attention, Sustained Attention, and Divided Attention, each with unique characteristics and applications.

4.2.5 Audio / Voice Analysis

Our "Data-Driven Monitoring Platform for Mental Health via Multi-modal Passive Sensing" incorporates the Audio-based MHW recognition Component as its central operations unit. The component uses speech as its foundation because it represents a common communication method among humans. Voice analysis within this system serves to detect user emotions thus revealing important health information about mental wellness. Figure 8 presents the nonintrusive benefits of using voice data analysis to detect MDD symptoms effectively.

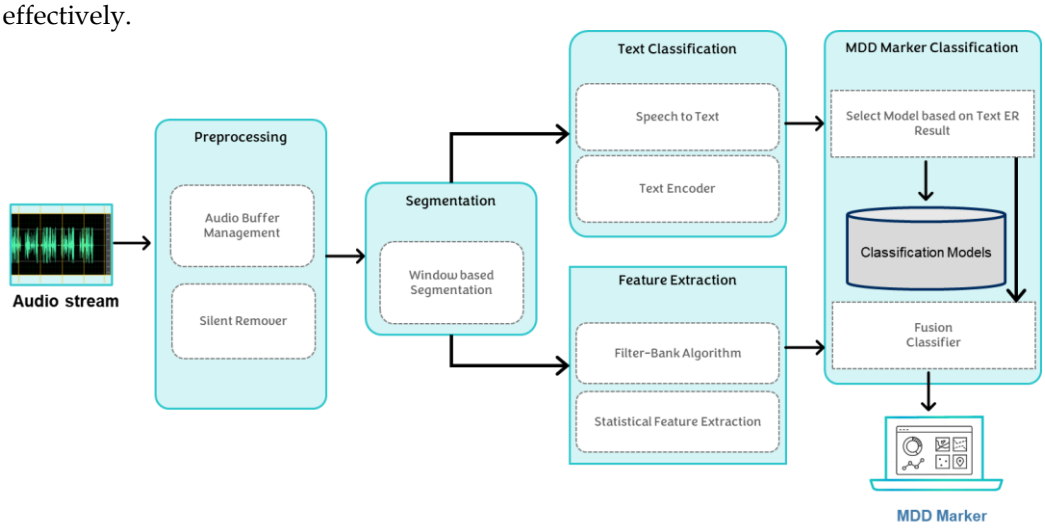

Figure 6. Sample MHW Workflow – MDD Detection Case using Audio Analysis.

The main objective of the Audio-based MHW recognition component relies on emotion recognition through subject voice data analysis. Voice Analysis measures human emotions through voice data analysis which occurs when users interact with the system using microphones. The model training process applies machine-learning models to process audio information. Model training forms the core of emotion detection through automated algorithms that analyze voice data in order to identify anger alongside happiness and neutral emotions and sadness and other emotions. The Non-Speech Reduction

techniques assist in differentiating between genuine vocal data and silent intervals and background sounds.

4.2.6 Triangulation in MDD

In the MDD detection case, triangulation will be a qualitative research strategy to test the validity of MDD detection and diagnosis through the convergence of information from multiple modes and sources of information such as self-reports, caregiver reports, clinical histories and evaluations, physiological reports, social media posts, voice analysis, video analysis, gaze analysis, and gestures of the subject. As an instance, in the case of MDD detection, the analysis of text messages posted by the subject while time-based sequential processing of messages may not always yield desirable results. It is because clusters of posts could contain irrelevant information. In addition, users with MHW issues may exhibit sudden shifts in mood, resulting in intervals of positive affect within their post sequences. To address these challenges, we can introduce various approaches for processing sets of user posts dedicated to detection of depression.

4.3 Analytics Layer

Real-Time Analytics evaluates MDD related data instantly to identify real-time behavioral patterns through observing changes in communication frequency with additional measurements of facial expressions and voice tone and physiological responses. The Behavioral Analytics module would analyze different behavioral aspects of diverse stakeholders through understanding their behaviors while also examining their preferences along with expectations and needs among patients with MDD and their caregivers and healthcare professionals. The distribution of users into separate groups according to demographic traits and behavioral patterns and specific requirements supports customized content creation and personalized service delivery. Through its Descriptive Analytics module the system would deliver insights into user emotional states alongside behavioral patterns and MDD trends and interaction data. The analytics system draws conclusions from available multi-modal data and text alongside images and audio as well as continuous physiological signals measured alongside platform usage patterns. Future MDD risks receive predictions by analyzing combined historical data with user behaviors and multichannel information inputs through the Predictive Analytics module.

This would facilitate anticipating and predicting any potential onset of MDD and enable proactive support and early intervention. In addition, multi-modal data on behavioral patterns and emotional trends would identify potential MDD triggers, patterns, and risk factors associated with the subject. Consequently, an early warning system on the risk of MDD would enable timely intervention.

4.4 Interface Layer

The interface layer forms the frontline of interaction enabling the ease with which users can navigate, engage with, and comprehend the mechanisms of a given system. In the MDD domain, the three interface sub-layers provide ease of interaction and navigation as well as the requisite assurances MDD patients regarding security and privacy. The Interaction Sub-layer consists of API, SDK, UI, and FB. These modules cater to the interaction needs of various stakeholders like application designers, developers, users, and maintainers.

An important aspect of trust in MDD applications are privacy, security and ownership of the data and Eco-SMART is equipped with security in data on rest and data in transit. The security is across all the sublayers in horizontal queue. Consequently, the data security aware protection is active at every stage of data acquisition, storage, analytics, and usage.

4.5 Services Layer

Users and clients access services through a security management layer which enforces rules and performs authentication followed by authorization for access. The layer first verifies authorized rights along with permission before performing exact operations. The platform implements sensitive data management while reviewing MDD/MHW app security with the Eco-SMART system through its core platform structural implementation of privacy and security measures. It would provide a set of well-defined and reusable services that can be accessed by various client applications or components. This would free up the MHW app developers from worrying about issues unrelated to the specific domain, promoting modular and decoupled architecture as well as facilitating maintenance, testing, and scalability.

5. Future Directions

The seminal Eco-SMART framework provides a multitude of benefits. This framework is expected to spur further research and development in various directions like app development, framework development, specialized LLMs in MHW, interoperability standards in MHW apps, etc. Here we discuss some of the most promising future directions.

The Eco-SMART framework will encourage authors as well as various research groups and businesses to implement it and put it into practice. In addition, Eco-SMART is expected to encourage the development of similar platforms in various other healthcare related domains like infectious diseases and behavioral therapies. More specifically, we plan to enhance the Eco-SMART framework by employing an agentic paradigm for covering desirable features such as agency, autonomy, contextualization, empowerment, etc. Eco-SMART should also invite researchers and practitioners in the LLM area to create specialized LLMs for the use in MHW applications.

The availability of a unified, accessible, and scalable cloud-computing platform like Eco-SMART would not only spur the development of the MHW apps but also help multiple apps seamlessly coordinate/integrate with each other to provide a fuller spectrum of MHW related services in prevention, detection, diagnosis, prognosis, treatment, and rehabilitation/support. Moreover, the availability of such unified and scalable platforms in MHW would promote the development of interoperability standards across healthcare systems in various jurisdictions, IT applications, etc. Such efforts would enable addressing challenges like fragmented systems, non-standardized data elements, privacy concerns, and workflow integration in MHW practices can successfully overcome such obstacles and realize significant improvements in information exchange, integrated care, patient outcomes in a wider context.

6. Conclusion

Mental health and wellness relate to the emotional, psychological, and social well-being of individuals, which has become a global challenge to societies. These challenges are costing trillions of dollars in socio-economic losses by creating social impediments, financial burdens, and individual disabilities, highlighting the importance of technological advancements in detection, treatment, and management mental health issues. The relative dearth of unified and scalable platforms for supporting MHW apps hinders the employment of such promising technologies as artificial intelligence and real-time big data analytics. To bridge this gap, we have proposed an encompassing framework for building such a platform offering a variety of requisite services and capabilities to MHW app developers and users. These capabilities include multimodal MHW data acquisition, data processing, secure data storage, data authentication, intelligent analytics, maker detection/interpretation, and other APIs. Such a unified, scalable, intelligent, and adaptable platform simplifies the process of connecting an app with these services, enabling MHW

app developers to focus on building the core functionality rather than dealing with complex backend infrastructure and interoperability of various MHW apps. In addition, this initiative should push the growth of MHW apps by providing effective and productive app design and development services and user support.

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